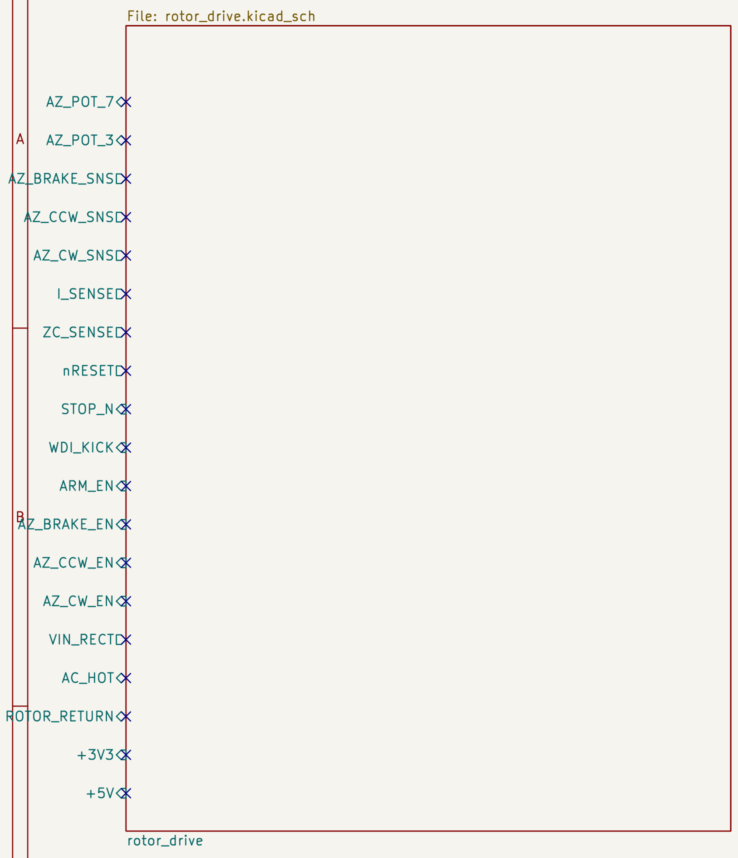

F407 rotator controller
Root sheet. This PR is the rotor interface and the AC motor drive only.
The processor sheet and the power sheet are not drawn yet.
No-connects on these sheet pins are the waiver for that.
Power arrives later on +5V and +3V3. PWR_FLAG on the child sheet marks that.
Do not copy hardware/controller/ (imported V4). It stays as reference.

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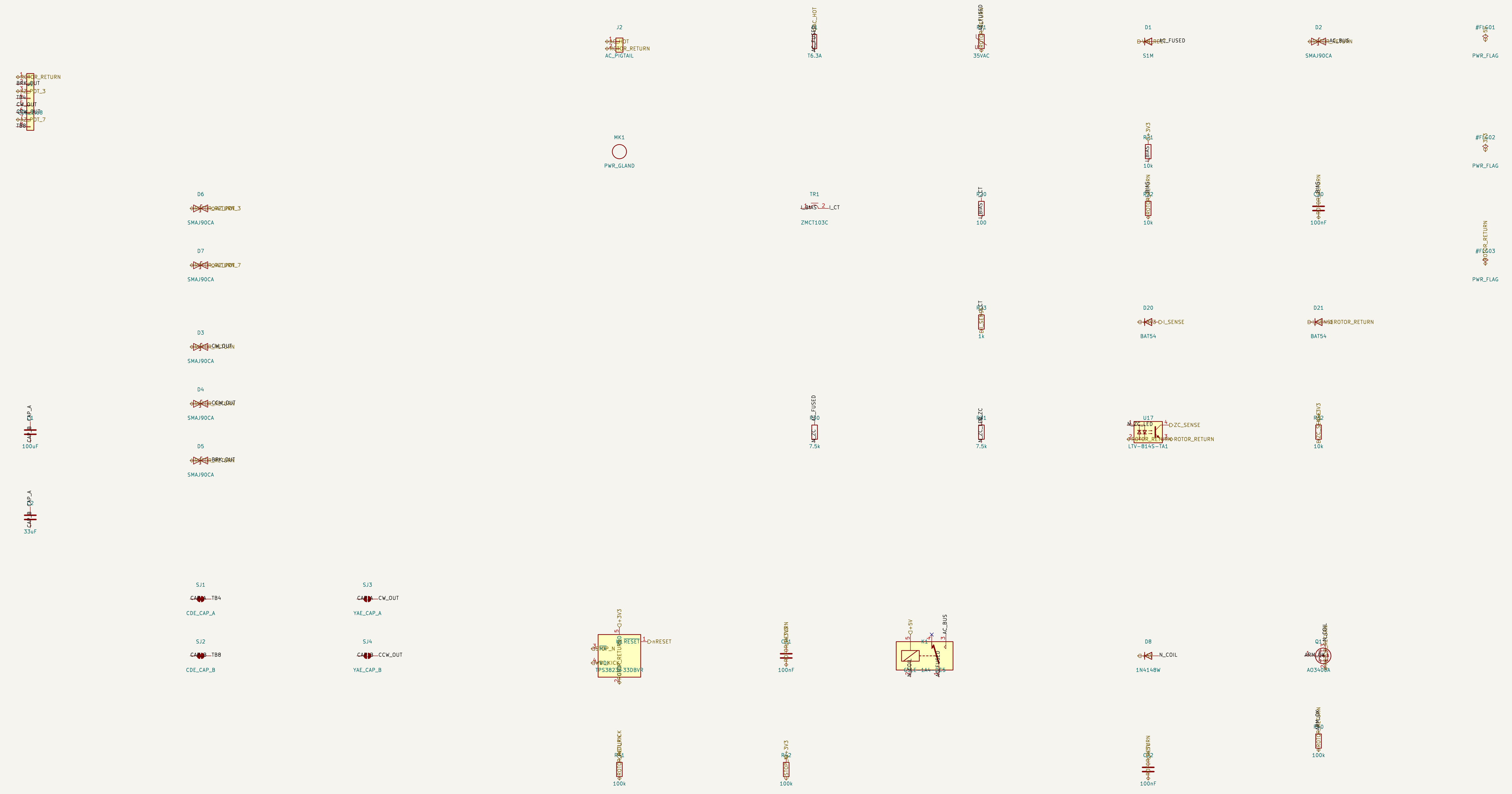
ROTOR INTERFACE AND AC MOTOR DRIVE (azimuth, AC version)
F407 controller, first sheet. MCU pin assignment is deferred.
Logic ground is ROTOR_RETURN (CDE terminal 1). See DESIGN-NOTES.md and ADR 0018.
Hierarchical labels leave this sheet for the processor sheet and the power sheet.

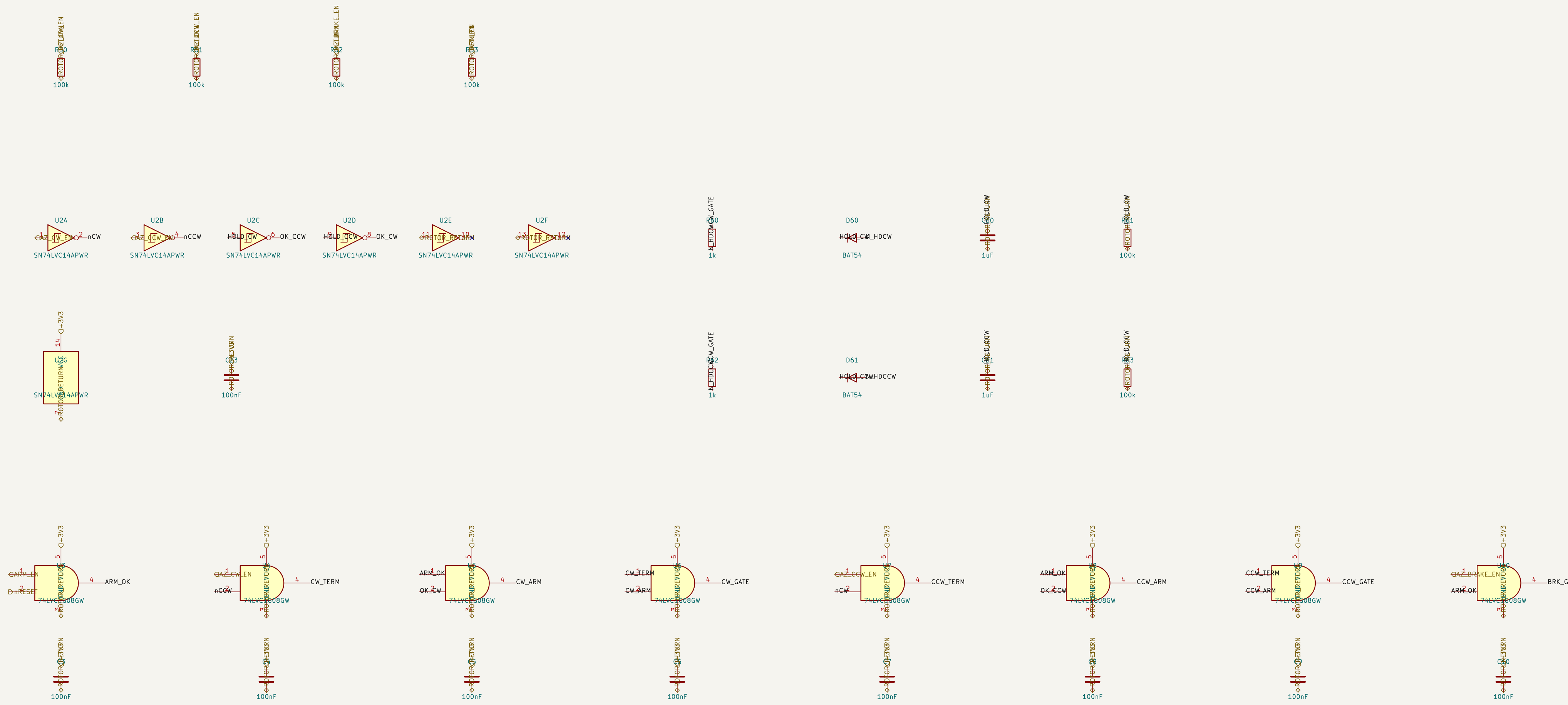
CDE terminal block J1, 1:1 with the rotor strip (catalog hygain-ham-iv, V4 sheet note).
1 ROTOR_RETURN common: motor, brake, pot wiper
2 BRAKE   solenoid. Energized = released. Off = wedge engaged
3 AZ_POT_3   pot end +
4 CAP / winding end. Run cap across 4 and 8 for CDE (SJ1, SJ2 bridged)
5 CW   via the right limit switch to terminal 8
6 CCW   via the left limit switch to terminal 4
7 AZ_POT_7   pot end −
8 CAP / winding end
V4's J3 is a 10-pin Mate-N-Lok. Its pin numbers are not these terminal numbers.
The V4 text note (1 Return, 2 Brake, 4 Cap Left, 5 Motor Right, 6 Motor Left, 8 Cap Right)
matches this map. Right = CW, Left = CCW.

Yaesu AC (G-450 class): land motor CW (Yaesu 5) on J1-5, motor CCW (Yaesu 4) on J1-6,
motor common (Yaesu 6) and pot ground (Yaesu 3) on J1-1, pot REF (Yaesu 1) on J1-3,
pot wiper (Yaesu 2) on J1-7. Open SJ1/SJ2, close SJ3/SJ4 so the cap sits across the drive ends.
G-5500 and HDR-300 keep the cap in the rotor: leave SJ1-SJ4 open and do not fit C1/C2.
HDR-300's pinout is not this 1:1 map. It needs a cable adapter. Cap omitted either way.
Brake enable AZ_BRAKE_EN high RELEASES the brake. Any fault or ARM drop engages it.

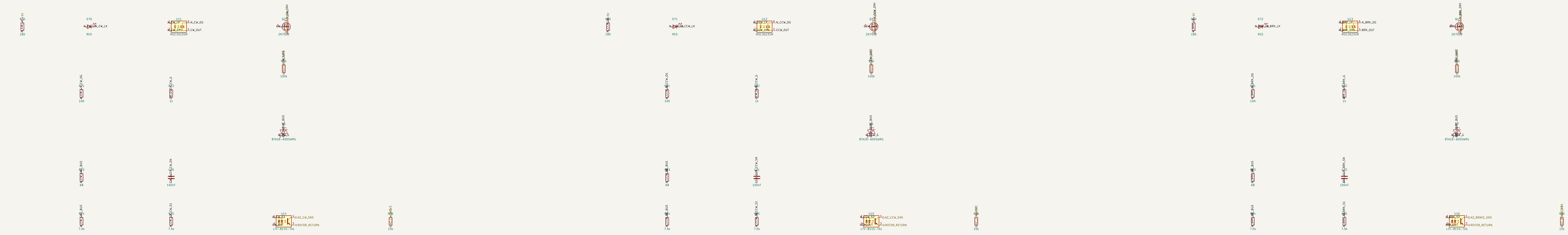
26 VAC inlet, fuse, MOV, half-wave tap, current transformer, ARM relay. Logic stays up when ARM opens.



Interlock, 3.3 V. CW_GATE = CW_EN AND NOT CCW_EN AND ARM_OK AND NOT HOLD_CCW.
ARM_OK = ARM_EN AND nRESET. BRK_GATE = AZ_BRAKE_EN AND ARM_OK.
HOLD is about 70–110 ms (1 uF, 100k, 74LVC14). See DESIGN-NOTES.md.



Each channel, left to right: 180 ohm from +5V, red LED (anode toward the resistor), MOC3023SM,
2N7002 to ROTOR_RETURN. Below: 100 ohm gate, 1k gate-to-MT1, BTA16-600SW, 68 ohm + 0.1 uF X2,
LTV-814 stuck-on sense. Triacs stand vertical on the bar, 12.7 mm pitch, lead row near board x 34.4.



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